Experiment 9: Bag-of-Visual-Words (BoVW) for Image Classification

Theory:
The Bag-of-Visual-Words (BoVW) model is a popular technique used in computer vision for image classification. It draws inspiration from the Bag-of-Words model used in text classification, where an image is treated as a document and local features (such as LBP and HOG) are treated as visual words. The key steps include feature extraction from local patches of images, clustering of these features to form a visual vocabulary, representing each image as a histogram of visual word occurrences, and then classifying using a machine learning model such as SVM.

Algorithm Steps:
1. Load and preprocess the dataset (e.g., CIFAR-10).
2. For each image, extract descriptors using techniques like:
   a. Local Binary Patterns (LBP)
   b. Histogram of Oriented Gradients (HOG)
3. Flatten and concatenate the descriptors for each image.
4. Apply KMeans clustering to all descriptors to create a visual vocabulary.
5. Represent each image as a histogram of cluster occurrences (visual words).
6. Train a Support Vector Machine (SVM) classifier on the BoVW histograms.
7. Evaluate the classifier on the test set using metrics like accuracy and confusion matrix.

Flowchart (written format):
Start → Load CIFAR-10 Dataset → Select Subset of Images → For Each Image:
→ Convert to Grayscale → Extract LBP Features → Extract HOG Features → Concatenate Descriptors
→ Combine Descriptors from All Images → Apply KMeans Clustering → Form Visual Vocabulary
→ For Each Image: Predict Cluster and Create Histogram → Use Histogram as BoVW Representation
→ Train SVM Classifier → Predict on Test Images → Evaluate Results → End